# Atharva Chougule

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## Education

**Stanford University** — *Sept 2025 – Present*
*MS in Computer Science (Systems specialization)*

- GPA: **4.13/4**

**Indian Institute of Technology Madras** — *July 2019 – May 2023*
*Bachelor of Technology with Honors in Computer Science and Engineering*

- GPA: **9.79/10** | Class Rank: **1**
- **Gold Medalist** for being the student with **highest GPA** in Computer Science Department.

## Research Experience

**Compiling Sparse Tensor Algebra for GPUs and Multicore CPUs** — *Stanford University*
*Stanford Compiler Lab | Advisor: Prof. Fredrik Kjolstad* — *Aug 2025 - Present*

- Designed the **first partitioning algorithm that provably load-balances** any sparse tensor algebra expression across parallel execution units.
- Implemented the algorithm in the **Nacho compiler**, generating sparse kernels for **multi-core CPUs and GPUs**.
- Nacho-generated kernels are on par with or faster than **Intel MKL and cuSPARSE** (geomean 0.93–5.5×), **TACO** (1.2–2.4×), and **PyTorch Sparse** (2.0–9.1×).
- Currently working on **unifying automatic load balancing, vectorization, and tiling** for sparse tensor kernels.

**Code Generation for Distributed Graph Algorithms** — *IIT Madras*
*Bachelor's Thesis | Advisor: Prof. Rupesh Nasre* — *Dec 2022 - May 2023*

- Architected and built a **compiler** for the **distributed systems backend** of **StarPlat** - a DSL for graph analytics capable of generating code for multiple target architectures (**GPUs, multi-core and distributed systems**).
- Designed new **DSL primitives** and **graph representation** for representing **distributed dynamic graphs**.
- Generated **MPI code** for graph algorithms like Triangle Counting, Page Rank, Betweenness Centrality and SSSP.

**Evaluating Byzantine Fault Tolerance for a new model of distributed computing** — *IIT Madras*
*Research Project | Advisor: Prof. John Augustine* — *Aug 2022 - Nov 2022*

- Modeled a new framework for **distributed computing** where, unlike the traditional MPC model, **data is decoupled from the distributed network** and is stored in an external entity like the Cloud.
- Studied **Byzantine Fault Tolerance** in the new model and developed a **Conversion theorem** from classic **MPC** algorithms.

## Publications

**Partitioning Unstructured Sparse Tensor Algebra for Load-Balanced Parallel Execution** — Paper | Code
**Atharva Chougule***, Alexander J Root*, Rubens Lacouture, Bobby Yan, Rohan Yadav, Fredrik Kjolstad
***ASPLOS '27*** *(To appear)*

**Generating Morph Algorithms for Multiple Backends** — Paper | Code
Ashwina Kumar, Nibedita Behera, **Atharva Chougule**, Mohammed Shan, Rushabh Lalwani, Rupesh Nasre
*Under Review*

## Work Experience

**Waymo LLC** — *Mountain View, CA*
*Software Engineer Intern | ML Compute Team* — *June 2026 – Present*

- Built a **PJRT** based **JAX runtime** backend for **Waymo's CarTPU chips**, enabling **JIT compilation** and execution of arbitrary custom JAX models on the platform.
- **Extended JAX with inline assembly support**, enabling **ML hardware architects and performance engineers** to combine high-level JAX with hand-optimized assembly kernels for **hardware-specific performance tuning**.

**Rubrik, Inc** — *Bangalore, India*
*Software Engineer | Rubrik Security Cloud Private (RSC-P) - Platform Team* — *June 2023 - April 2025*

- Engineered a **distributed and horizontally scalable architecture** for Rubrik Security Cloud Private (RSC-P) using **Kubernetes**. This resolved the compute bottleneck for RSC-P's job framework, increasing its supported scale by more than **5x**.
- Developed a **state machine framework** for RSC-P upgrades to make cluster upgrades robust and predictable.
- Built **defragmentation tools** for the **MySQL database** resulting in **storage reduction by more than 70%**.

*Software Engineer 2 | Rubrik Security Cloud Private (RSC-P) - Platform Team* *April 2025 - July 2025*

- Led a project for hardening **security and data resilience** of RSC-P by **enforcing backups and automating disaster recovery**.
- Improved system stability by optimizing resource utilization and addressing security gaps resulting in more than **50% decrease in customer-reported issues.**

**Rubrik, Inc** *Bangalore, India*
*Software Engineer Intern | Rubrik Cloud Vault(RCV)* *May 2022 – July 2022*

- Designed microservices-based framework to sync customer cost metrics for **Rubrik Cloud Vault (RCV)**. Built a general framework to sync Cost data from **Azure Cost API** to Rubrik's Database and trigger alerts for unusual cost patterns.

## Invited Talks

**Partitioning Unstructured Sparse Tensor Algebra for Load-Balanced Parallel Execution**

- 2026 Global Open Compute Project (OCP) Summit *October 2026*
- PORTAL Retreat *August 2026*
- Stanford Software Lunch *April 2026*

## Scholastic Achievements and Honors

- Awarded the **B. Ravichandran Memorial Prize** for being the student with **best academic performance** and **highest GPA** in the Computer Science and Engineering Department. *2023*
- Nominated for **Engineering Excellence Award** at Rubrik. *2024*
- Awarded the prestigious **Kishore Vaigyanik Protsahan Yojana (KVPY)** fellowship *2019*

## Teaching

**CS143: Compilers** *Spring 2026*
*Teaching Assistant under Prof. Fredrik Kjolstad* *Stanford University*

- Developed coursework, ran office hours, graded assignments, and managed course logistics for the Compilers course.

## Mentoring

- **Mentoring a visiting ETH Zurich researcher** on her master's thesis at Stanford on parallelizing scatter-based sparse tensor algebra kernels.
- At Rubrik, **mentored an intern** on his project and guided new graduates' transition to industry through the **Rubrik Mentorship Program**.
- **Mentored high-school students** preparing for **JEE Main and JEE Advanced**, India's national engineering entrance exams.

## Selected Academic Projects

**TensorDSL for rPi-QPU** *Spring 2026 | Code*
*Course Project: CS240LX - Advanced Systems Laboratory | Instructor: Prof. Dawson Engler* *Stanford University*

- Designed and implemented a **DSL for Array programming** for the **Broadcom VideoCore GPU** on **r-Pi**.
- Built a Compiler to lower simple high-level Tensor Programs to **QASM assembly**.

**Speaker-Microphone Audio Bootloader** *Winter 2026 | Code*
*Course Project: CS140E - Operating Systems Implementation | Instructor: Prof. Dawson Engler* *Stanford University*

- Implemented an **audio bootloader** for Raspberry Pi to bootload and run any programs on the Pi using audio.
- **Implemented custom drivers** for a speaker and microphone to get **maximum audio-bandwidth** for the bootloader.

**Compiler for MacroJava** *Autumn 2022 | Code*
*Course Project: Compiler Design | Instructor: Prof. Kartik Nagar* *IIT Madras*

- Developed a compiler for a subset of Java, supporting **conditionals, loops, control sequences, classes, etc**.
- Implemented a lexical analyser and parser using **Flex** and **Bison**. Translated the generated AST to **microIR**, which was further translated to **miniRA for register allocation** and was finally translated to the **MIPS Assembly**.